$A \otimes B = \begin{bmatrix} A_{11}\cdot B & A_{12}\cdot B & \cdots \\ A_{21}\cdot B & \cdots & \\ \vdots & & A_{mn}\cdot B \end{bmatrix}$

$A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A \otimes B = \begin{bmatrix} 0\cdot B & 2\cdot B \\ 0\cdot B & 0\cdot B \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$A \otimes B \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0\cdot B & 2\cdot B \\ 0\cdot B & 0\cdot B \end{bmatrix} = \begin{bmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

		0	1
	0	1	0
0	-1	0	-1
	-2	-1	-2
	0	1	0
1	-1	0	-1
	-2	-1	-2

3x2

down a block: add row dim. of B

right a block: sub. col dim of B

-5: 1, 0 block -2 diag

-4: 1, 0 " -1 diag

-3: 1, 0 " 0 diag, 1, 1 block -2 diag


A B
C D
[A B C D]